

Kubernetes Application Troubleshooting Guide (Production Level)

1. Pod Not Starting (CrashLoopBackOff)

- Check pod status: `kubectl get pods`
- Describe pod: `kubectl describe pod`
- Check logs: `kubectl logs`
- Common causes: wrong image, missing env variables, DB connection failure, wrong command/entrypoint.

2. Pod Stuck in Pending

- Check node resources: `kubectl describe node`
- Check taints/tolerations.
- Check PVC binding issues.
- Common causes: insufficient CPU/memory, storage class misconfiguration.

3. Service Not Accessible

- Verify service: `kubectl get svc`
- Check service selector matches pod labels.
- Test inside cluster: `kubectl exec -it -- curl`
- Common causes: selector mismatch, wrong targetPort, network policy blocking traffic.

4. Ingress Not Working

- Check ingress resource: `kubectl get ingress`
- Check ingress controller pod status.
- Verify DNS resolution.
- Common causes: incorrect host/path rules, TLS secret missing.

5. Database Connection Failure

- Verify DB pod running.
- Check service DNS: nslookup
- Check network policies.
- Validate credentials from secret.

6. ConfigMap/Secret Updated But App Not Reflecting

- Pods do not restart automatically on ConfigMap update.
- Restart deployment: `kubectl rollout restart deployment`
- Best practice: use config hash annotation.

7. High Latency / Slow Application

- Check resource usage: `kubectl top pod`
- Check HPA status.
- Investigate database performance.
- Check for throttling or CPU limits.

8. Random Pod Restarts

- Check describe pod for OOMKilled.
- Verify liveness/readiness probes.
- Check resource limits configuration.

9. Rolling Update Failure

- Check rollout status: `kubectl rollout status deployment`
- Review events: `kubectl get events`
- Rollback if needed: `kubectl rollout undo deployment`

10. Production Debug Flow (Step-by-Step)

- Step 1: Is pod running?
- Step 2: Can service reach pod?
- Step 3: Is DNS resolving?
- Step 4: Are network policies blocking?
- Step 5: Check logs and metrics.
- Step 6: Verify external dependencies (DB, cache, API).